Introduction to Internet of Things

Assignment-Week 4

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 15 **Total marks: 15 X 1= 15**

QUESTION 1:

State whether the following statement is true or false.

Statement: Coverage in WSN is defined as the entire area is covered satisfactorily.

a. True
b. False

Correct Answer: b. False

Detailed Solution: Coverage in WSN is defined as the area-of-interest is covered satisfactorily.

See lecture 17 (Sensor Networks-IV) @ 01:44

QUESTION 2:

State True or False.

Statement: "In Wireless Sensor Network coverage, we study how to deploy or activate sensors to cover the monitoring area."

a. True
b. False

Correct Answer: a. True

Detailed Solution: In Wireless Sensor Network coverage, we study how to deploy or activate sensors to cover the monitoring area.

See lecture 17 (Sensor Networks-IV) @ 02:00

QUESTION 3:

Water reservoir overflow monitoring in WSN is an example of which type of reporting?

a. Event driven
b. On-demand
c. Scattered
d. None of these

Correct Answer: a. Event driven

Detailed Solution: Water reservoir overflow monitoring in WSN is an example of event driven reporting.

See lecture 17 (Sensor Networks-IV) @ 08:13

QUESTION 4:

State whether the following statement is true or false.

Statement: Objective of coverage in WSN is to use minimum number of sensors and minimize network lifetime.

a. True

b. False

Correct Answer: b. False

Detailed Solution: The objective of coverage in WSN is to use minimum number of sensors and maximize the network lifetime.

See lecture 17 (Sensor Networks-IV) @ 08:47

QUESTION 5:

State whether the following statement is true or false.

Statement: "Sensor nodes in WSN can be of two types – 1. Static sensor nodes, and 2. Mobile sensor nodes."

a. True

b. False

Correct Answer: a. True

Detailed Solution: Sensor nodes in WSN can be of only two types – 1. Static sensor nodes, and 2. Mobile sensor nodes.

See lecture 17 (Sensor Networks-IV) @ 02:00

QUESTION 6:

An inventory control system is an example of _____?

a. Event-driven reporting
b. On-demand reporting
c. No sensing and reporting involvement
d. None of these

Correct Answer: b. On-demand reporting

Detailed Solution: An inventory control system is an example of On-demand reporting.

See lecture 17 (Sensor Networks-V) @ 08:40

QUESTION 7:

Most problems in static WSN can be classified as -

a. Area coverage
b. Barrier coverage
c. Both (a) and (b)
d. Neither (a) nor (b)

Correct Answer: c. Both (a) and (b)

Detailed Solution: Most problems in static WSN can be classified as -

a. **Area coverage**

b. **Point coverage**

c. **Barrier coverage**

See lecture 17 (Sensor Networks-V) @ 11:30

QUESTION 8:

What are the disadvantages of Stationary WSN?

a. Node failure may result in the partition of networks
b. Topology cannot be changed automatically
c. Both (a) and (b)
d. Neither (a) nor (b)

Correct Answer: c. Both (a) and (b)

Detailed Solution: The disadvantages of Stationary WSN are -

a. Node failure may result in the partition of networks
b. Topology cannot be changed automatically

See lecture 18 (Sensor Networks-V) @ 00:52

QUESTION 9:

State True or False.

Statement: "MANET-infrastructure less network of mobile devices connected wirelessly in WSN follow the self-HOP properties."

a. True
b. False

Correct Answer: a. True

Detailed Solution: MANET-infrastructure-less network of mobile devices connected wirelessly in WSN follow the self-HOP properties.

See lecture 18 (Sensor Networks-V) @ 04:44

QUESTION 10:

In Mobile WSN, the Data Mules

a. Collect the data from sensor nodes
b. Goes to the sink and delivers the collected data
c. Both (a) and (b)
d. Neither (a) nor (b)

Correct Answer: c. Both (a) and (b)

Detailed Solution: In Mobile WSN, the Data Mules

a. Collect the data from sensor nodes
b. Goes to the sink and delivers the collected data

See lecture 18 (Sensor Networks-V) @ 07:15

QUESTION 11:

_____ senses different under the sea or water level parameters.

a. UAV
b. AUV
c. Both (a) and (b)
d. Neither (a) nor (b)

Correct Answer: b. Autonomous Underwater Vehicle (AUV)

Detailed Solution: AUV senses different under the sea or water level parameters.

See lecture 18 (Sensor Networks-V) @ 08:35

QUESTION 12:

Humans carry their devices and move around. Sensors embedded within the devices record readings. Sensory readings are then transmitted for processing.

This paradigm of sensing is known as –

a. Machine Centric Sensing
b. Device Centric Sensing
c. Human Centric Sensing
d. None of these

Correct Answer: c. Human Centric Sensing

Detailed Solution: Humans carry their devices and move around. Sensors embedded within the devices record readings. Sensory readings are then transmitted for processing.
This paradigm of sensing is known as Human Centric Sensing.

See lecture 18 (Sensor Networks-V) @ 11:37

QUESTION 13:

State True or False.

Statement: "Energy of Devices and Participant selection are two major problems in Human Centric Sensing."

a. True
b. False

Correct Answer: a. True

Detailed Solution: The following are the challenges in human-centric sensing -

a. Energy of devices
b. Participant selection
c. Privacy of users

See lecture 18 (Sensor Networks-V) @ 12:45

QUESTION 14:

Which of the following network topologies is used in UAV networks?

a. Mesh
b. Star
c. Both (a) and (b)
d. Neither (a) nor (b)

Correct Answer: a. Both (a) and (b)

Detailed Solution: UAV networks use the mesh and star network topologies.

See lecture 19 (UAV Networks) @ 02:43

QUESTION 15:

The _____ provides integrated services based on device collected data-sets.

a. M2M Application Platform
b. M2M Service Platform
c. M2M Hardware Platform
d. None of these

Correct Answer: a. M2M Application Platform

Detailed Solution: The M2M Application Platform provides integrated services based on device collected data-sets.

See lecture 20 (Machine to Machine Communication) @ 18:00

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